

Water and Sanitation Through Time

Teacher Answer Key



Key Enquiry Questions

- How much progress did the Romans make in public health?
- Why did public health decline during the Middle Ages?
- To what extent did towns become filthier during the Early Modern period?
- How did the Industrial Revolution lead to a public health crisis?
- Why did it take the 'Great Stink' to finally clean up Britain's streets?

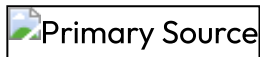
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Class:

How much progress did the Romans make in public health?

Q1. How does this source demonstrate both the advancements and limitations of Roman public health?

Source A: Roman communal latrines at Housesteads Fort (Hadrian's Wall)



***What is this source showing?** This is a well-preserved example of a Roman communal latrine block. Notice the stone seating with holes positioned over a deep channel where running water continuously flushed waste away. Below the seats is a shallower gutter that carried clean water, used by the Romans to wet a communal sponge-on-a-stick (xylospongium) for wiping.*

Do Now Activity

Undefined. PAST TOPIC: What is a primary source?

A piece of evidence created at the time of the event being studied (e.g., a diary, photograph, or artefact).

Undefined. PAST TOPIC: What is a secondary source?

A piece of evidence created after the event by someone who was not there (e.g., a modern history book).

Undefined. PAST TOPIC: What does 'chronological order' mean?

Arranging events in the order in which they happened in time, from earliest to most recent.

Undefined. PAST TOPIC: What century is the year 1854 in?

The 19th Century.

Qundefined. PAST TOPIC: What is the difference between BC and AD (or BCE and CE)?

BC (Before Christ) refers to years before the birth of Jesus, while AD (Anno Domini) refers to years after. The numbers in BC count backwards, while AD count forwards.

Source B: Seneca the Younger (c. 4 BC – AD 65)

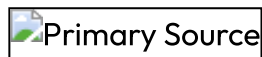


Who is this? *This is a bust of the Roman philosopher Seneca the Younger.*

Why did public health decline during the Middle Ages?

Q1. What does this highly detailed plumbing plan suggest about the role of monasteries in preserving public health during the Medieval era?

Source A: The Canterbury Cathedral Waterworks Plan (c. 1165)



***What is this source showing?** This extraordinary map, drawn around 1165, shows the complex water system built for the monks of Christ Church Priory in Canterbury. It details pipes drawing fresh water from springs outside the city walls, directing it to a settling tank, and then routing it through the monastery to washbasins, the kitchen, and finally flushing the latrines.*

Do Now Activity

Qundefined. What was the main feature of Iron Age Settlements & Cesspits?

Farming roundhouses built near natural water sources with simple, temporary earth cesspits located away from drinking water.

Qundefined. How did Roman Conduits & Water Supply impact society?

They brought massive quantities of clean spring water over miles using gravity, powering public bathhouses, fountains, and flushing latrines.

Qundefined. Identify one major problem with The Roman Withdrawal.

The advanced centralized sanitation infrastructure was abandoned, causing public hygiene to decline back into primitive, localized methods.

Qundefined. Why were Iron Age cesspits safe?

Because populations were small and spread out.

Qundefined. How did the Romans move water?

Using gravity through stone conduits over many miles.

Qundefined. What were Roman latrines?

Communal public toilets flushed continuously by running water.

Qundefined. What happened to sanitation when the Romans left?

The systems collapsed and towns reverted to primitive methods.

Qundefined. Did Iron Age people pollute their water?

No, they intentionally dug cesspits far from their drinking water.

Qundefined. Why was flowing water important to Romans?

They believed it was vital for keeping society healthy and clean.

Qundefined. What is a conduit?

A channel or pipe for conveying water.

To what extent did towns become filthier during the Early Modern period?

Q1. Why do you think this brilliant invention failed to catch on in Early Modern Britain despite its obvious sanitary benefits?

Source A: Sir John Harington's flushing toilet design (1596)



***What is this source showing?** This diagram is from a 1596 pamphlet by Sir John Harington, a godson of Queen Elizabeth I. It illustrates his invention: the first modern flushing water closet, which he called the 'Ajax'. It featured a flush valve to release water and a wash-down design to empty the bowl.*

Do Now Activity

Qundefined. What was the main feature of Peasant Cesspits?

Simple garden outhouses dug over earth holes. Safe in low-density rural villages like Wharram Percy, but highly problematic in crowded towns.

Qundefined. How did Monastic luxury (Canterbury Priory) impact society?

It showcased advanced water engineering (lead pipes, color-coded fresh and waste water lines) funded by wealthy, organized Christian orders.

Qundefined. Identify one major problem with Gongfermers and night-work.

It was a dangerous, filthy task limited to nighttime, which only temporarily cleared overflowing urban cesspits without solving the lack of permanent sewers.

Qundefined. What was a Gongfermer?

A medieval worker who cleared human waste from cesspits at night.

Qundefined. Why did medieval monks build water systems?

They believed cleanliness brought them closer to God.

Qundefined. What did Edward III order in 1357?

The removal of filth from London's streets to prevent sickness.

Qundefined. Why were medieval towns dangerous for health?

Overcrowding caused cesspits to leak into shared drinking wells.

Qundefined. How did the Romans keep forts clean?

Using deep sewers flushed continuously with running water.

Qundefined. How did water sellers make money?

By selling barrels of river water to townspeople.

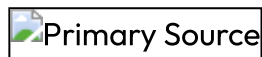
Qundefined. What is Miasma?

A bad smell previously thought to cause disease.

How did the Industrial Revolution lead to a public health crisis?

Q1. How did Dr. Snow use this map to challenge the prevailing 'miasma' theory of disease?

Source A: Dr. John Snow's Cholera Map of Soho (1854)



What is this source showing? *This famous map was created by Dr. John Snow during the devastating 1854 cholera outbreak in Soho, London. Each black bar on the map represents a death from cholera at that specific address. Snow noticed that the deaths clustered tightly around one specific public water pump on Broad Street.*

Chronological Big Picture

1750: Rural Britain - *Most of the population lives and works in the countryside.*

1780s: First Factories - *Steam-powered textile mills begin mass production.*

1800-1850: Rapid Urbanisation - *Millions migrate to industrial towns looking for work.*

1831: First Cholera Epidemic - *Cholera arrives in Britain, killing thousands.*

1842: Chadwick's Report - *Edwin Chadwick publishes his damning report on sanitary conditions.*

Undefined. Why do you think disease spread so quickly in the newly built industrial towns?

Why did it take the 'Great Stink' to finally clean up Britain's streets?

Q1. What does the sheer scale of this brickwork reveal about the Victorian government's response to the Great Stink of 1858?

Source A: Construction of Joseph Bazalgette's intercepting sewers



***What is this source showing?** This photograph captures the colossal engineering effort required to build the London sewer system in the 1860s. Designed by Joseph Bazalgette, the project involved constructing 82 miles of massive enclosed brick intercepting sewers and 1,100 miles of street sewers to stop sewage from flowing directly into the River Thames.*

Do Now Activity

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Overcrowding caused cesspits to leak into shared drinking wells.

Qundefined. What terrifying waterborne disease first struck Britain in 1831?

Cholera.

Qundefined. What scientific theory did doctors wrongly believe caused disease before 1860?

Miasma Theory (bad smells).

Qundefined. Who published a damning report on sanitary conditions in 1842?

Edwin Chadwick.

Qundefined. What was the 'laissez-faire' attitude?

The government's belief that it should not interfere in public health or people's daily lives.

End of Unit Quiz Pack

Instructions: Answer the 50 quick-fire recall questions below. If you get stuck, the scrambled answers are provided in the Answer Bank on the final page.

1. What technology did Roman engineers use to bring fresh water over miles into British towns?

2. Where sat Roman soldiers side-by-side over water-flushed latrine channels?

3. What did Roman soldiers use to wipe themselves in communal latrines?

4. What did Roman bathers experience inside a bathhouse like Bearsden?

5. What is a famous Roman bath complex located in Somerset, England?

6. What is a 'hypocaust' in Roman architecture?

7. What were Roman public toilets called?

8. What Roman philosopher complained about the noise in a public bathhouse?

9. How did early Iron Age Britons dispose of waste safely?

10. Why was Iron Age waste disposal effective?

11. Why did medieval villages like Wharram Percy avoid major public health crises despite using simple cesspits?

12. On the twelfth-century plans of Canterbury Priory, what did the red lines represent?

13. How did poorer townspeople get their fresh water if they did not own a private well?

15. Why did medieval 'gongfarmers' perform their job exclusively under the cover of darkness?

17. Where did garderobes in castles usually empty their waste?

19. Who was employed to clean out medieval cesspits?

21. Why did London's sanitation problems get worse during the Early Modern period?

23. Who invented the first water closet (flushing toilet) in Britain in 1596?

25. What was a 'water seller's' job in an Early Modern town?

27. In what year was the New River completed?

14. What material did archaeologists discover in medieval cesspits that shows how villagers wiped themselves?

16. What was the term for a medieval toilet, meaning 'wardrobe'?

18. What organization in medieval times had the best access to clean piped water?

20. What was the main source of water for early medieval towns before pipes were common?

22. Which famous seventeenth-century Londoner wrote about his cellar being flooded by his neighbor's toilet waste?

24. Why was the 1596 flushing toilet not used by ordinary citizens?

26. Who constructed the New River to bring fresh water to London?

28. What were Early Modern 'conduits'?

29. Why did Samuel Pepys' cellar flood with waste?

30. What was a major consequence of overcrowded towns in the Early Modern period?

31. By how much did Britain's population grow during the century of industrialisation (1750–1850)?

32. How did working-class families living in terraced yards usually obtain their water?

33. What was the terrifying waterborne disease that first struck Britain in 1831, killing 31,000 people?

34. What scientific theory did Victorian doctors believe in before they understood that cholera was spread by dirty water?

35. Why did cesspits under shared yard privies frequently overflow in industrial cities?

36. What was a 'back-to-back' house?

37. Who published the 1842 Report on the Sanitary Condition of the Labouring Population?

38. What act was passed in 1848 as a result of Chadwick's report?

39. What was the main problem with the 1848 Public Health Act?

40. What was the primary symptom of Cholera?

41. How did John Snow stop the Soho cholera outbreak of 1854?

42. What event in the summer of 1858 finally forced politicians to fund London's sewer network?

43. Which French scientist proved 'Germ Theory' in 1860, showing that microscopic organisms cause disease?

44. What did the landmark 1875 Public Health Act force local councils to do?

45. How many miles of brick sewers did Joseph Bazalgette construct beneath London?

46. What was Joseph Bazalgette's profession?

47. What modern facility replaced the old, unhygienic shared privies by the late 19th century?

48. What happened to the River Thames during the Great Stink?

49. Why did Parliament suddenly care about the Great Stink?

50. How did John Snow map the 1854 Cholera outbreak?

Quiz Pack Answer Bank

1,300 miles • 1613 • A wet sponge attached to the end of a shared wooden stick • An underfloor heating system used in bathhouses • Aquae Sulis (Bath) • By digging simple cesspits in spread-out roundhouse settlements • Cesspits overflowed and contaminated drinking water supplies • Cheap, crowded industrial housing with shared privies and no back gardens • Cholera • Civil Engineer • Directly into the moat or a cesspit at the base of the wall • Edwin Chadwick • Ensure all houses had piped clean water and proper sewer connections • Flushing toilets connected to Bazalgette's sewer system • From a shared pump in the street or yard that only ran for a few hours • Garderobe • Gongfarmers (or Gong Farmers) • He mapped the deaths and noticed they clustered around the Broad Street pump • He persuaded the local parish to remove the Broad Street pump handle • His neighbor's cesspit overflowed directly into it • Houses lacked running piped water and connections to street sewers • Housesteads Fort on Hadrian's Wall • It skyrocketed from 6 million to 21 million • It was not compulsory, so many towns ignored it to save money • Landlords refused to pay for them to be emptied, and they were not connected to sewers • Latrines • Louis Pasteur • Low population density meant waste didn't contaminate water supplies • Miasma Theory (the belief that disease is spread by bad smells) • Monasteries and Priors • Pipes carrying dirty waste water away to flush the toilets • Public water fountains where citizens could collect fresh water • Rivers, streams, and local wells • Samuel Pepys • Seneca the Younger • Severe diarrhea, dehydration, and blue skin • Sir Hugh Myddelton • Sir John Harington • Stone channels called conduits that utilized the natural pull of gravity • The Great Stink of the River Thames • The low population density meant waste did not build up enough to contaminate water supplies • The population grew rapidly, putting too much pressure on crowded town systems • The Public Health Act 1848 • The river level dropped during a heatwave, exposing rotting sewage • The smell was so terrible it reached the Houses of Parliament • They bought it from water sellers who hauled river water in barrels • To avoid disrupting the busy town streets with terrible smells and waste carts • Transporting river water in large barrels on horseback to sell to homes • Washing in sequence through cold, warm, and hot rooms • Wild moss